KENTUCKY COMMUNITY & TECHNICAL COLLEGE SYSTEM

ESSENTIAL SKILL 8 · FROM YOUR COLLEAGUES

Earth's Timeline Challenge

Geology · Relative and Absolute Dating

Small groups work asynchronously to solve a geologic mystery. They read rock layers, fossils, and radiometric data, build a timeline of events, and present their conclusions in a shared presentation or infographic.

Contributed by **Sherri Basil** and adapted with permission for *Designing Collaboration That Works*, the KCTCS faculty guide to Essential Skill 8. Student-facing sections are written to copy into Blackboard. Draft for review, in-house resource.

COURSE
Physical geology

LENGTH
About one week

TEAM SIZE
3–5 students, with assigned roles

FORMAT
Asynchronous, online

FINAL PRODUCT
Shared presentation, infographic, or narrated video, plus individual reflections

Closest level in the guide: **Level 3 · Coordinated**

1 · Quick | 2 · Structured | **3 · Coordinated** | 4 · Sustained

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Overview

FOR FACULTY

Working asynchronously in small groups, students will investigate a geologic mystery by analyzing rock layers, fossils, and numerical age data. The team will collaborate to determine the sequence of geologic events using relative dating principles and then compare those findings with absolute dating evidence. Students will create a shared presentation or infographic explaining their conclusions.

Learning Objective

By the end of this activity, students will be able to:

- Differentiate between relative and absolute dating methods.
- Apply geologic principles to determine the sequence of Earth events.
- Interpret numerical age data from radiometric dating.
- Collaborate effectively to analyze evidence and communicate scientific conclusions.

Sample Announcement

FOR STUDENTS

Earth's Timeline Challenge Begins This Week!

This week you will work with your team to investigate a geologic mystery. Using rock layers, fossils, geologic structures, and radiometric dating data, your group will reconstruct Earth's history and determine the sequence of geologic events. Be sure to contribute to your group discussion, support your conclusions with evidence, and respond to at least two other groups' findings.

Remember: geologists rarely work alone. Collaboration and evidence-based reasoning are essential parts of scientific investigation.

Student Instructions

FOR STUDENTS

Step 1: Join Your Team

You will be assigned to a group of 3-5 students.

Each team member should assume one of the following roles:

- Geologic Evidence Specialist
- Fossil Analyst
- Radiometric Dating Specialist
- Timeline Coordinator
- Presentation Manager

Step 2: Analyze the Evidence

Review the materials provided by your instructor, which may include:

- Rock layer diagrams
- Fossil records
- Cross-cutting relationships
- Intrusions and fault data
- Radiometric dating results

As a group, discuss: (see the Group Discussion Prompt)

1. Which rock layers are oldest and youngest?
2. What clues help establish relative age?
3. What does the radiometric dating data reveal?
4. How does absolute dating support or challenge the relative dating sequence?

Record your observations in your group's shared document.

Step 3: Create a Geologic Timeline

Using your evidence:

- Arrange events in chronological order.
- Label relative dating evidence used.
- Include numerical ages where provided.
- Identify at least one uncertainty or limitation in the evidence.

Step 4: Develop a Collaborative Presentation

Create a 5-7 slide presentation, infographic, or narrated video that includes:

1. Introduction to the geologic site
2. Explanation of relative dating principles used
3. Explanation of absolute dating methods used
4. Final timeline of events
5. Team conclusion and justification

Step 5: Peer Collaboration Reflection

After submitting the project, each student will post a brief reflection discussing:

- Their contribution to the group
- What they learned about Earth's history
- How collaboration improved the investigation

Minimum: 150 words

Group Discussion Prompt

FOR STUDENTS

As a group, review the geologic data provided by your instructor and work together to interpret the sequence of events recorded in the rock layers.

In your discussion, address each of the following questions:

1. **Which rock layers appear to be the oldest and which appear to be the youngest?**
   - Explain your reasoning using evidence from the rock sequence.
   - Consider the Law of Superposition and any features that may affect the order of layers.
2. **What clues help establish the relative age of the rock layers?**
   - Discuss the evidence used to determine the order of events.
   - Consider fossils, cross-cutting relationships, inclusions, unconformities, faulting, or intrusive igneous bodies.
3. **What does the radiometric dating data reveal?**
   - Examine the numerical ages provided.
   - What geologic events or rocks can be assigned actual ages?
   - How do these dates help interpret Earth's history?
4. **How does the absolute dating data support or challenge the relative dating sequence?**
   - Compare the timeline developed from relative dating evidence with the radiometric ages.
   - Do the two methods agree?
   - If there are differences, propose possible explanations.

Group Response Requirements

As a group, create one discussion post that includes:

- A description of the oldest and youngest rock layers.
- At least three pieces of evidence used for relative dating.
- An interpretation of the radiometric dating results.
- A conclusion explaining how relative and absolute dating work together to reconstruct geologic history.

Discussion Goal

By completing this discussion, you will apply the vocabulary and concepts of physical geology, use scientific evidence to explain Earth's history, and collaborate with your team to build a geologic timeline. These skills support the course learning outcomes of using geologic terminology, explaining Earth processes, and applying scientific reasoning.